

Analog Cruise Control System for DC Motor Applications

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Description

A DC Motor Driver and Controller will be designed, built and tested. The driver will control the power supplied to a small motor based on signals from the controller. The controller will use PWM to maintain a desired speed.

Inputs

- Speed Setpoint
- Supply Power

Output

- Rotating motor shaft

Specifications

- The driver will supply a minimum of 1 amp to a motor.
- The system will use a PI controller.
- Motor speed will be within $\pm 10\%$ of setpoint at steady state from 50 to 150 RPM.
- The system will maintain motor speed despite reasonable change in load from 50 to 150 RPM.
- The system will use an encoder to measure speed of the motor shaft.

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